1 Algebraic laws for nearness and convergence

Source: `PrimeTensor/Analysis/Near/Laws.lean`

What this file establishes

`PrimeTensor/Analysis/Limit.lean` introduced nearness and convergence on MulReal but related them to nothing: neither the multiplication of `PrimeTensor/Algebra.lean` nor its inversion was known to interact with them. This file supplies the four compatibilities, and one consequence.

nearness	multiplication	costs one level
nearness	inversion	costs nothing
convergence	multiplication	costs one level
convergence	inversion	costs nothing

The consequence is that ratios of convergent sequences converge to the ratio of the limits. Together the four are the sequential statement that the carrier operations are continuous.

Nothing new is estimated. The two nearness laws transport scaleWithin_mul and scaleWithin_inv of `PrimeTensor/Algebra.lean` across the quotient, and the two convergence laws are the anchor-combining pattern used three times already. What the file does show, for the first time, is that the deliberately weak nearness relation of `PrimeTensor/Analysis/Limit.lean` is strong enough to be transported at all — see Remark 1.3.

1.1 Nearness respects the operations

Theorem 1.1 (`MulReal.scaleNear_mul`)**.** *If*

$$\mathrm{ScaleNear}(\mathsf{succ}\,\ell,\ a,\ a') \quad \textit{and} \quad \mathrm{ScaleNear}(\mathsf{succ}\,\ell,\ b,\ b'),$$

then $\mathrm{ScaleNear}(\ell,\ ab,\ a'b')$*.*

```
theorem scaleNear_mul {level : Depth} {a a' b b' : MulReal}
    (ha : ScaleNear (.succ level) a a')
    (hb : ScaleNear (.succ level) b b') :
    ScaleNear level (a * b) (a' * b') := by
  obtain ⟨as, as', has, has', aAnchor, haTail⟩ := ha
  obtain ⟨bs, bs', hbs, hbs', bAnchor, hbTail⟩ := hb

  refine ⟨MulCauchyStream.mul as bs,
    MulCauchyStream.mul as' bs', ?_, ?_, ?_⟩

  · rw [← has, ← hbs]
    rfl

  · rw [← has', ← hbs']
    rfl
```

Proof. Each hypothesis supplies a pair of representative streams and an anchor: say (α, α') with anchor c_a for the first, and (β, β') with anchor c_b for the second. Offer as witnesses for the conclusion the pointwise products $\alpha\beta$ and $\alpha'\beta'$, using the stream multiplication of `PrimeTensor/Algebra.lean`.

The two identifications are `rfl` after rewriting by the given ones: MulReal's multiplication was defined by lifting the stream multiplication along the quotient, so the class of a product of streams is by construction the product of their classes, with no lemma needed.

For the tail, take $\mathrm{join}(c_a, c_b)$. For n at or after it, the two bounds on join and transitivity of AtOrAfter, all from `PrimeTensor/Completion.lean`, put n in both original tails, so both factor pairs are ScaleWithin(`succ` ℓ)-close at n; and scaleWithin_mul of `PrimeTensor/Algebra.lean` combines them into ScaleWithin(ℓ)-closeness of the products. ☐

Theorem 1.2 (`MulReal.scaleNear_inv`)**.** ScaleNear(ℓ, a, b) *implies* ScaleNear(ℓ, a^{-1}, b^{-1}).

Proof. Invert the two witnessing streams pointwise; the identifications are again `rfl`, because inversion on MulReal was also defined by lifting. Keep the anchor and the level, and apply scaleWithin_inv termwise. ☐

Remark 1.3 (The existential form is what makes this work). ScaleNear was defined in `PrimeTensor/Analysis/Limit.lean` by quantifying *existentially* over representatives, and that file's design note observed that the choice makes the relation weaker and easier to establish. Here it is what makes it transportable. Both proofs above proceed by *constructing* witnesses for the conclusion out of the witnesses supplied by the hypotheses, and the construction is available because the operations act pointwise on streams and because MulReal's operations were defined as their lifts — so the required identifications hold definitionally rather than by an argument.

Under a universally quantified nearness the same statements would have to be proved for representatives one does not get to choose, and each proof would have to bridge from an arbitrary representative to a convenient one, spending levels. So the two theorems above would not have the level costs they do.

1.2 Convergence respects the operations

Theorem 1.4 (`MulReal.converges_mul` and `converges_inv`)**.** *If $a \to x$ and $b \to y$ then $a_n b_n \to xy$; and if $a \to x$ then $a_n^{-1} \to x^{-1}$.*

```
theorem converges_mul {a b : Seq} {x y : MulReal}
    (ha : ConvergesTo a x)
    (hb : ConvergesTo b y) :
    ConvergesTo (fun n => a n * b n) (x * y) := by
  intro level
  obtain ⟨aAnchor, haTail⟩ := ha (.succ level)
  obtain ⟨bAnchor, hbTail⟩ := hb (.succ level)
```

Proof. For the product, fix a level ℓ, instantiate both hypotheses at `succ` ℓ, take join of the two anchors, and for n beyond it apply Theorem 1.1 to the two nearness facts. For the inverse, instantiate at ℓ, keep the anchor, and apply Theorem 1.2 pointwise. ☐

Corollary 1.5 (`MulReal.converges_ratio`)**.** *If $a \to x$ and $b \to y$ then* $\mathrm{ratio}(a_n, b_n) \to \mathrm{ratio}(x, y)$.

Proof. Unfold ratio to $a_n \cdot b_n^{-1}$ and compose the two theorems. ☐

Design note 1.6 (One motif, five layers). The pair "multiplication costs one level, inversion costs none, and combining two hypotheses needs a join of anchors" has now been proved five times, at every layer the development has:

Layer	Multiplication	Inversion
MulRat, single pair	scaleWithin_mul	scaleWithin_inv
Cauchy streams	MulCauchyStream.mul	MulCauchyStream.inv
asymptotic equivalence	MulAsymptotic.mul	MulAsymptotic.inv
MulReal, nearness	Thm. 1.1	Thm. 1.2
MulReal, convergence	Thm. 1.4	

The first three are in `PrimeTensor/Algebra.lean` and the last two here. The repetition is not redundancy: each layer's statement is about different objects, and only the last two could be stated at all before this file. But the *reason* for the level cost is the same at every layer, and it is the one identified in `PrimeTensor/Algebra.lean` — multiplication combines two independent errors and their ratios multiply, whereas inversion merely reverses orientation and multiplicative distance does not notice. The cost is a property of the operation, not of the layer, so it propagates unchanged all the way up.

Remark 1.7 (This is continuity, sequentially). Theorem 1.4 is the statement that multiplication and inversion are continuous, in the sequential form available here: they carry convergent sequences to convergent sequences with the expected limits. Since `PrimeTensor/Analysis/Limit.lean` defined convergence for sequences and not through a filter or a topology, this is the strongest form the statement can take in the present vocabulary. In particular nothing here asserts joint continuity in any uniform sense, and nothing asserts that the limit is unique — that gap, open since `PrimeTensor/Structure.lean`, is untouched, so "the" limit still may not be spoken of.

Remark 1.8 (What is not proved). Nearness is still not transitive at any level, for the reason `PrimeTensor/Analysis/Limit.lean` gave, and nothing here changes that. There is no compatibility of nearness or convergence with ofRat, no statement that a convergent sequence is Cauchy in any sense — the vocabulary for that is still missing — and no interaction with the differential notions of `PrimeTensor/Analysis/Derivative.lean`: in particular NegligibleRelative is not shown to be preserved by products, which is the fact a product rule would need.

Summary of the correspondence

Lean declaration	Kind	Here
`MulReal.scaleNear_mul`	theorem	Thm. 1.1
`MulReal.scaleNear_inv`	theorem	Thm. 1.2
`MulReal.converges_mul`	theorem	Thm. 1.4
`MulReal.converges_inv`	theorem	Thm. 1.4
`MulReal.converges_ratio`	theorem	Cor. 1.5

Every declaration of `PrimeTensor/Analysis/Near/Laws.lean` appears above. The file contains no `sorry`, no axiom, and no unproved named proposition. Design note 1.6 and Remarks 1.3, 1.7 and 1.8 are stated for the reader and are not declarations of the file.